

# Supplementary Material for “GaussFM: Compact Foundation-Feature Gaussian Memory for Open-Vocabulary 3D Scene Understanding”

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## I. OVERVIEW

This supplementary material records additional calibration, protocol controls, and qualitative examples for the main paper. The main paper keeps the narrative focused on the deployed compact field: rendered-view LERF-OVS grounding, LERF-OVS direct 3D object selection, VALA-aligned ScanNet direct point querying, and frame-wise RADIO comparisons. The tables below provide the supporting controls behind those claims.

Unless otherwise stated, all LERF-OVS numbers are evaluated on the same four annotated scenes used in the main paper. Direct 3D object-selection controls render selected Gaussian primitives only for mask evaluation; the text query is applied before rendering.

## II. RENDERED-VIEW BOUNDARY CALIBRATION

Table I expands the rendered-view mask calibration summarized in the main paper. The fixed threshold and peak-connected-component mask conversion is label-free and uses the heatmap argmax only to remove disconnected regions far from the queried peak.

## III. DIRECT 3D PROTOCOL CONTROLS

Tables II–V document the registration and confidence controls used to understand direct 3D primitive querying. Multiview Primitive Registration is treated as a registered supervision and analysis bridge, while the main compact-memory query in the paper uses Gaussian-center primitive scores with fixed support-calibrated selection and does not read a multiview-registration feature cache at inference.

TABLE I

LABEL-FREE RENDERED-MASK BOUNDARY CALIBRATION ON THE FIXED LERF-OVS CHECKPOINTS. THRESHOLD VARIANTS CHANGE ONLY THE PEAK-RELATIVE BINARY MASK THRESHOLD. THE FINAL VARIANT KEEPS THRESHOLD 0.60 AND ADDS PEAK-CONNECTED-COMPONENT FILTERING, WHICH REMOVES DISCONNECTED MASK REGIONS AWAY FROM THE QUERY HEATMAP MAXIMUM. LOCAcc IS MEASURED FROM THE HEATMAP ARGMAX BEFORE MASK EXTRACTION.

| Readout                     | Fig.         | Ramen        | Tea.         | Waldo        | Macro        |
|-----------------------------|--------------|--------------|--------------|--------------|--------------|
| Threshold 0.50              | <b>0.431</b> | 0.586        | 0.549        | 0.411        | 0.494        |
| Threshold 0.55              | 0.430        | 0.611        | 0.568        | 0.451        | 0.515        |
| Threshold 0.60              | 0.424        | <b>0.620</b> | 0.576        | 0.477        | 0.524        |
| Threshold 0.65              | 0.415        | 0.619        | 0.578        | 0.490        | 0.525        |
| Threshold 0.70              | 0.395        | 0.596        | 0.556        | 0.477        | 0.506        |
| Threshold 0.60 + peak comp. | <b>0.513</b> | <b>0.625</b> | <b>0.618</b> | <b>0.527</b> | <b>0.571</b> |

TABLE II

MULTIVIEW PRIMITIVE REGISTRATION CONTROLS FOR LERF-OVS DIRECT 3D OBJECT SELECTION. EXCEPT FOR THE EXPLICITLY MARKED EARLIER RAW-CENTER VARIANT, RESULTS USE THE SAME FIXED TOP-2% PRIMITIVE SELECTOR. "REG." IS THE MEAN FRACTION OF GAUSSIANS RECEIVING REGISTERED RENDERED-VIEW FEATURES; UNREGISTERED PRIMITIVES USE THE DIRECT GAUSSIAN-CENTER FALLBACK UNLESS DISABLED BY THE ABLATION.

| Variant                      | Selector / views      | Reg.  | mIoU  | Acc@0.25 |
|------------------------------|-----------------------|-------|-------|----------|
| Raw Gaussian-center          | top10%, earlier       | –     | 0.080 | 0.093    |
| Raw Gaussian-center          | top2%, MPR protocol   | –     | 0.012 | 0.009    |
| MPR, cosine                  | top2%, all poses 24   | 0.385 | 0.312 | 0.547    |
| MPR, softmax                 | top2%, all poses 24   | 0.385 | 0.342 | 0.555    |
| MPR + voxel                  | top2%, train views 24 | 0.369 | 0.349 | 0.595    |
| MPR + voxel                  | top2%, all poses 8    | 0.240 | 0.277 | 0.487    |
| MPR + voxel                  | top2%, all poses 48   | 0.448 | 0.380 | 0.653    |
| MPR + voxel                  | top2%, all poses 96   | 0.485 | 0.385 | 0.643    |
| MPR + voxel, lower blend     | top2%, all poses 96   | 0.485 | 0.385 | 0.631    |
| MPR + voxel, w/o calibration | top10%, all poses 24  | 0.385 | 0.070 | 0.099    |
| MPR + voxel, w/o depth check | top2%, all poses 24   | 0.972 | 0.347 | 0.615    |

TABLE III

MULTIVIEW-REGISTRATION-TO-FIELD CONSISTENCY FOR LERF-OVS DIRECT 3D OBJECT SELECTION. ALL VARIANTS USE THE SAME SOFTMAX-THRESHOLD SELECTOR FAMILY, 0.5% FLOOR, 1.8% CAP, AND LABEL-FREE COLOR-EDGE BOUNDARY CALIBRATION. RAW AND REGISTRATION-TO-FIELD VARIANTS REPORT PER-SCENE SWEEP-BEST THRESHOLDS; REGISTERED MPR REPORTS THE FIXED `THR0p25` PAPER RESULT. REGISTRATION-TO-FIELD DISTILLS REGISTERED SUMMARY TARGETS INTO THE COMPACT DIRECT FIELD AND EVALUATES DIRECT GAUSSIAN-CENTER SCORING WITH A 0.5 RESIDUAL BLEND TO THE ORIGINAL FIELD. THE CONFIDENCE-WEIGHTED VARIANT USES LABEL-FREE LOG VIEW-COUNT WEIGHTS FROM MULTIVIEW REGISTRATION.

| Readout                      | Metric   | Fig.  | Ramen | Tea.  | Waldo | Macro |
|------------------------------|----------|-------|-------|-------|-------|-------|
| Raw Gaussian-center          | mIoU     | 0.001 | 0.006 | 0.158 | 0.019 | 0.046 |
| MPR-to-field consistency     | mIoU     | 0.488 | 0.438 | 0.510 | 0.212 | 0.412 |
| MPR-to-field + conf. weights | mIoU     | 0.404 | 0.598 | 0.520 | 0.225 | 0.436 |
| Registered MPR control       | mIoU     | 0.531 | 0.580 | 0.566 | 0.243 | 0.480 |
| Raw Gaussian-center          | Acc@0.25 | 0.000 | 0.000 | 0.237 | 0.045 | 0.071 |
| MPR-to-field consistency     | Acc@0.25 | 0.714 | 0.578 | 0.695 | 0.364 | 0.588 |
| MPR-to-field + conf. weights | Acc@0.25 | 0.643 | 0.746 | 0.678 | 0.409 | 0.619 |
| Registered MPR control       | Acc@0.25 | 0.786 | 0.746 | 0.763 | 0.409 | 0.676 |

TABLE IV

DIRECT3D MECHANISM AUDIT USING LABEL-FREE MULTIVIEW-REGISTRATION COVERAGE, PRIMITIVE REGISTERED-SCORE CONFIDENCE, AND TEXT-MARGIN AMBIGUITY. SCENE RESULTS REPORT MEAN VALID REGISTERED VIEWS; SCORE-BUCKET RESULTS REPORT MEAN TOP-1% PRIMITIVE TEXT SCORE; TEXT-MARGIN RESULTS REPORT SCORE MARGIN AGAINST THE STRONGEST COMPETING CATEGORY.

| Group                   | Item          | Support | Coverage/score | mIoU   | Acc@0.25 |
|-------------------------|---------------|---------|----------------|--------|----------|
| Scene                   | figurines     | 56      | 9.5911         | 0.5309 | 0.7857   |
| Scene                   | ramen         | 71      | 20.8716        | 0.5805 | 0.7465   |
| Scene                   | teatime       | 59      | 13.0006        | 0.5662 | 0.7627   |
| Scene                   | waldo_kitchen | 22      | 5.7542         | 0.2429 | 0.4091   |
| Registered-score bucket | low           | 70      | 0.1800         | 0.4345 | 0.6143   |
| Registered-score bucket | mid           | 69      | 0.4257         | 0.5132 | 0.7101   |
| Registered-score bucket | high          | 69      | 0.6939         | 0.6358 | 0.8551   |
| Text margin             | ambiguous     | 70      | -0.0882        | 0.4329 | 0.6286   |
| Text margin             | mixed         | 69      | 0.2451         | 0.5303 | 0.7246   |
| Text margin             | distinct      | 69      | 0.6074         | 0.6202 | 0.8261   |

TABLE V

QUERY-LEVEL AUDIT FOR LERF DIRECT 3D OBJECT SELECTION. CONFIDENCE INTERVALS ARE BOOTSTRAP INTERVALS OVER QUERY MASKS FOR THE FIXED `THR0p25` SELECTOR.

| Scene         | Queries | mIoU  | Acc@0.25 | Zero pred. |
|---------------|---------|-------|----------|------------|
| Figurines     | 56      | 0.531 | 0.786    | 0.071      |
| Ramen         | 71      | 0.580 | 0.746    | 0.028      |
| Teatime       | 59      | 0.566 | 0.763    | 0.017      |
| Waldo Kitchen | 22      | 0.243 | 0.409    | 0.182      |

TABLE VI

NEAREST-VIEW FRAME-WISE RADIO CACHE BASELINE ON LERF-OVS. EACH ANNOTATED TARGET FRAME USES THE CLOSEST CACHED RADIO FEATURE MAP BY CAMERA-CENTER DISTANCE, EXCLUDING THE TARGET FRAME ITSELF; THE FEATURE MAP IS NOT WARPED INTO THE TARGET CAMERA.

| Scene         | LocAcc | mIoU   | N  | Dist.  |
|---------------|--------|--------|----|--------|
| Figurines     | 0.1786 | 0.0755 | 56 | 0.2831 |
| Ramen         | 0.2817 | 0.1740 | 71 | 0.5443 |
| Teatime       | 0.3559 | 0.2114 | 59 | 0.4266 |
| Waldo Kitchen | 0.2727 | 0.1570 | 22 | 0.5787 |
| Macro         | 0.2722 | 0.1545 | –  | 0.4582 |

TABLE VII

PER-GAUSSIAN 1280-D EXPLICIT RADIO-MEMORY BASELINE ON LERF-OVS. CACHED RADIO FEATURES ARE REGISTERED TO GAUSSIAN CENTERS, STORED AS FP16 VECTORS, RENDERED TO ANNOTATED VIEWS, AND EVALUATED WITH THE SAME FROZEN SIGLIP2 SCORER.

| Scene         | LocAcc | mIoU   | Reg. frac. | Storage MiB |
|---------------|--------|--------|------------|-------------|
| Figurines     | 0.6429 | 0.2729 | 0.1123     | 412.1       |
| Ramen         | 0.6056 | 0.4333 | 0.2835     | 934.3       |
| Teatime       | 0.5085 | 0.3541 | 0.2647     | 1123.4      |
| Waldo Kitchen | 0.5000 | 0.2125 | 0.1475     | 1688.8      |
| Macro         | 0.5642 | 0.3182 | 0.2020     | 1039.7      |

#### IV. FEATURE-MEMORY AND COMPRESSION CONTROLS

Tables VI–IX compare the compact field against simple feature-memory alternatives and relate feature reconstruction quality to downstream relevance. These controls support the central design choice of reconstructing compact foundation-space features rather than storing a raw high-dimensional RADIO vector per primitive.

#### V. BOUNDARY AND GEOMETRY CONTROLS

Tables X and XI record boundary-oriented metrics for the SAM3-box boundary control. This control is reported as a boundary-assisted diagnostic rather than the core compact direct selection.

#### VI. TRAINING AUDITABILITY

Table XII records checks for the training entry point. The paper artifact manifest and checksum file provide the corresponding source-level provenance for tables, figures, and result registries.

TABLE VIII

COMPRESSION RATIO VERSUS DOWNSTREAM mIoU. SAVING RATIO COMPARES DIRECT PER-GAUSSIAN 1280-D FP16 RADIO-FEATURE STORAGE WITH THE COMPACT CHECKPOINT FOOTPRINT; DOWNSTREAM METRICS USE THE FINAL RENDERED-VIEW AND DIRECT3D COMPACT-MEMORY QUERY PROTOCOLS.

| Scene         | Saving | Rendered mIoU | Direct3D mIoU | Direct3D Acc@0.25 |
|---------------|--------|---------------|---------------|-------------------|
| Figurines     | 1.74×  | 0.6429        | 0.5936        | 0.9286            |
| Ramen         | 3.00×  | 0.5378        | 0.3843        | 0.6338            |
| Teatime       | 3.32×  | 0.7609        | 0.7104        | 0.8983            |
| Waldo Kitchen | 4.04×  | 0.6576        | 0.4861        | 0.7727            |

TABLE IX

FEATURE RECONSTRUCTION ERROR VERSUS TEXT RELEVANCE ERROR. FEATURE ERROR IS 1— BEST VALIDATION DECODED COSINE FROM THE SCENE TRAINING LOG; TEXT RELEVANCE ERRORS ARE DERIVED FROM THE FROZEN RENDERED-GROUNDING QUERY PROTOCOL.

| Scene         | Feature err. | mIoU err. | LocAcc err. | Best cosine |
|---------------|--------------|-----------|-------------|-------------|
| Figurines     | 0.2382       | 0.5756    | 0.1786      | 0.7618      |
| Ramen         | 0.1624       | 0.3799    | 0.0986      | 0.8376      |
| Teatime       | 0.2005       | 0.4240    | 0.1017      | 0.7995      |
| Waldo Kitchen | 0.2209       | 0.5231    | 0.1364      | 0.7791      |

TABLE X

MEASURED BOUNDARY-ERROR DIAGNOSTICS FOR THE STRICT SAM3-BOX DIRECT3D ROW. BOUNDARY ERROR IS 1—BOUNDARY-F AND TRIMAP ERROR IS 1—TRIMAP IoU.

| Scene         | mIoU   | Boundary-F | Boundary err. | Trimap IoU | Trimap err. |
|---------------|--------|------------|---------------|------------|-------------|
| Figurines     | 0.6136 | 0.7116     | 0.2884        | 0.3986     | 0.6014      |
| Ramen         | 0.6409 | 0.7713     | 0.2287        | 0.4400     | 0.5600      |
| Teatime       | 0.6130 | 0.7255     | 0.2745        | 0.4626     | 0.5374      |
| Waldo Kitchen | 0.4142 | 0.4638     | 0.5362        | 0.2820     | 0.7180      |

TABLE XI

ALPHA/DEPTH DISCONTINUITY ALIGNMENT COVERAGE FOR THE STRICT SAM3-BOX DIRECT3D ROW. GEOMETRY RECORDS COUNT PER-QUERY ALPHA/DEPTH EDGE METRICS AND OVERLAYS.

| Scene         | Queries | Geometry records | mIoU   | Boundary-F |
|---------------|---------|------------------|--------|------------|
| Figurines     | 56      | 56               | 0.6136 | 0.7116     |
| Ramen         | 71      | 71               | 0.6409 | 0.7713     |
| Teatime       | 59      | 59               | 0.6130 | 0.7255     |
| Waldo Kitchen | 22      | 22               | 0.4142 | 0.4638     |

TABLE XII

AUDITABILITY CHECKS FOR THE TRAINING ENTRY POINT. STATUS ‘RISK’ MEANS THE GUARD EXISTS BUT NEEDS RELEASE HARDENING.

| Check                 | Status | Severity | Evidence                                                               |
|-----------------------|--------|----------|------------------------------------------------------------------------|
| script_size           | pass   | medium   | 3735 lines; threshold 4000                                             |
| run_manifest          | pass   | high     | present                                                                |
| split_resolution      | pass   | high     | train/val split tokens present                                         |
| trusted_checkpoint_io | pass   | high     | load_trusted_checkpoint referenced                                     |
| training_lock         | pass   | medium   | lock acquisition/release present                                       |
| raw_tensor_load_sites | pass   | medium   | 1 raw torch.load site behind or replaced by load_training_tensor_cache |

Reconstructed scene features vs. frame-wise RADIO across frozen-head tasks

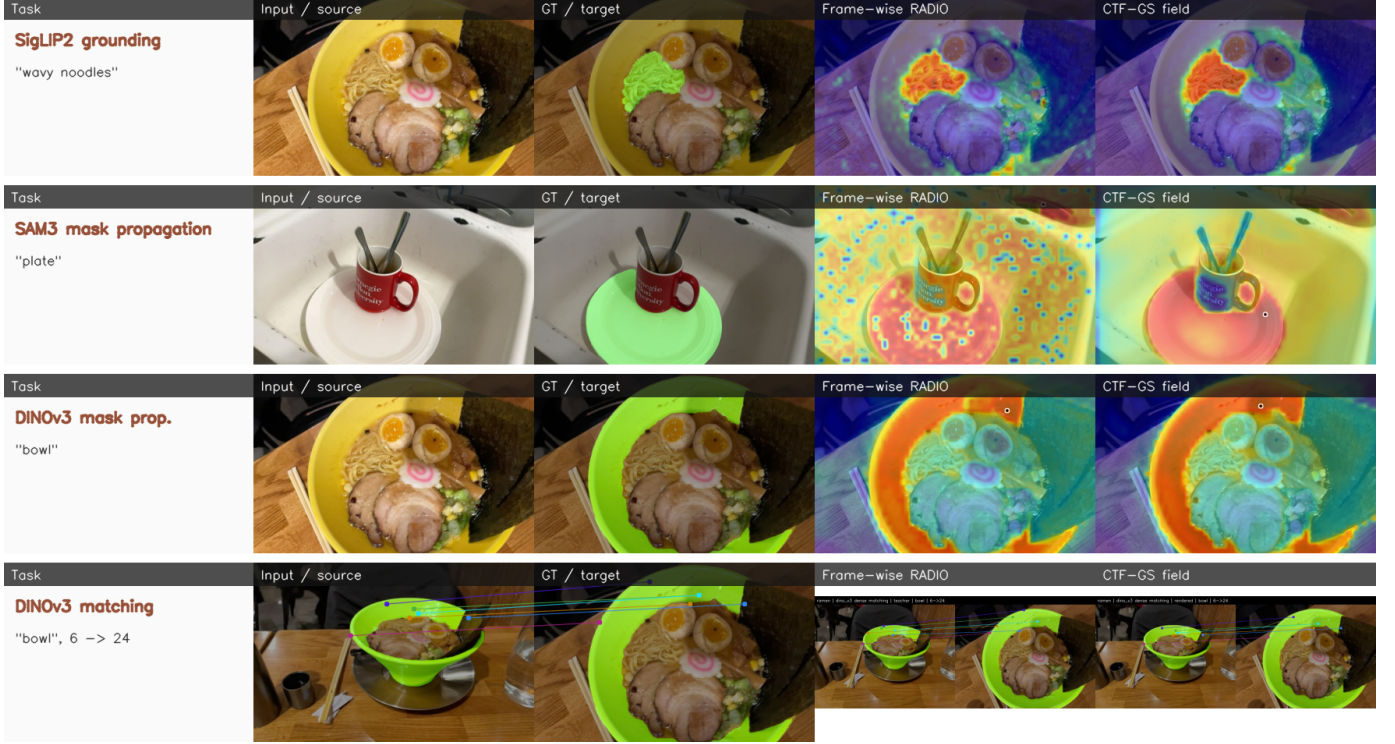


Fig. 1. Additional frozen-head qualitative examples. The SAM3-adaptor and DINOv3-adaptor panels are driven by reconstructed GaussFM features rather than an official RGB SAM decoder.

## VII. ADDITIONAL QUALITATIVE RESULTS

Figure 1 shows frozen-head adaptor examples. These examples are useful for interpreting the frame-wise foundation-feature comparison, but the main paper keeps the LERF qualitative figure focused on the cleaner 2D/3D open-vocabulary query taxonomy.

Figure 2 shows the registered multiview direct-3D control. It is included as an analysis path because Multiview Primitive Registration transfers registered multiview support to primitives, whereas the main compact direct selection uses the deployed compact Gaussian-center field.

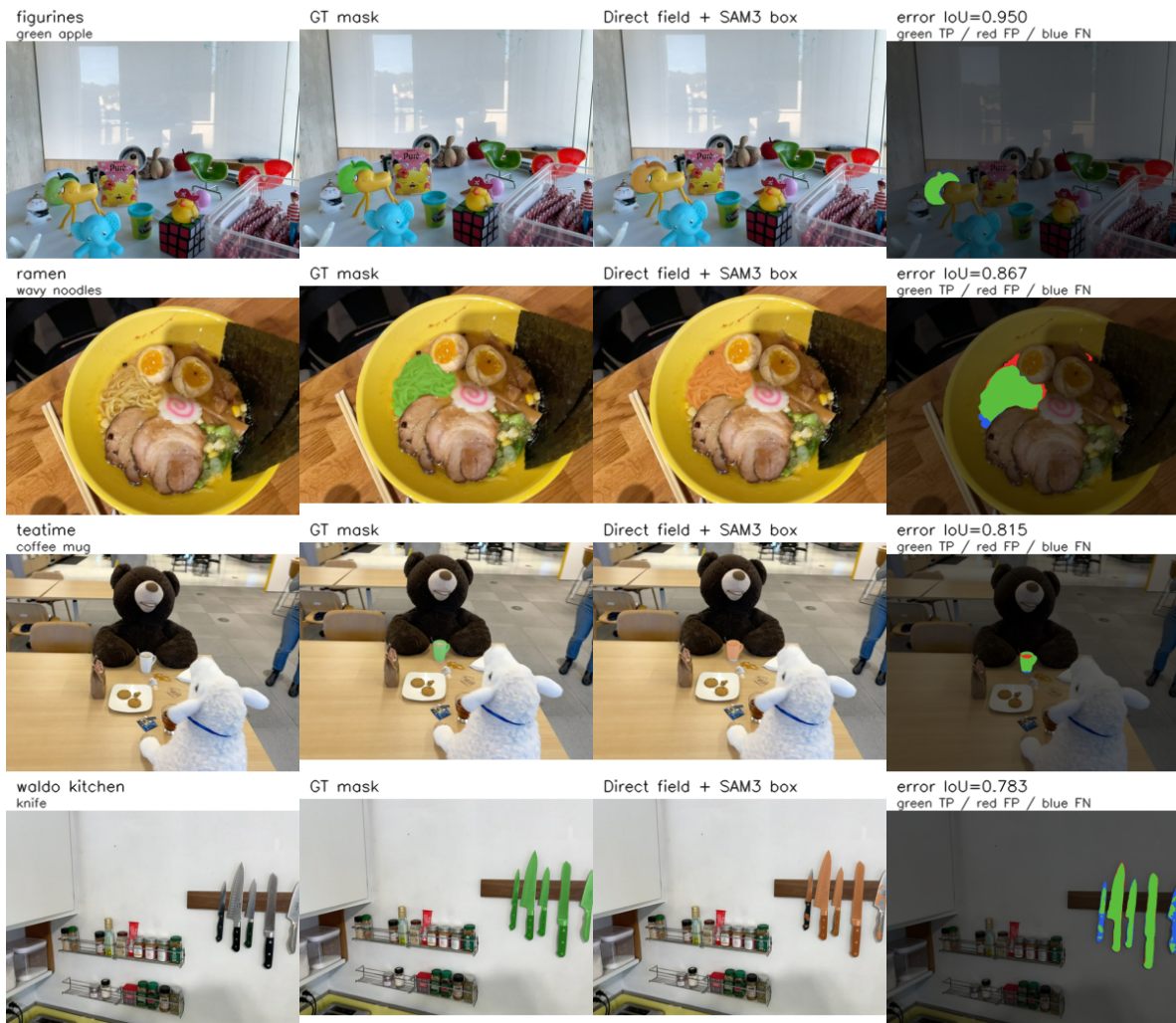


Fig. 2. Registered multiview direct-3D control examples. Multiview Primitive Registration is used as an analysis and training bridge, not as the main compact-field inference cache.



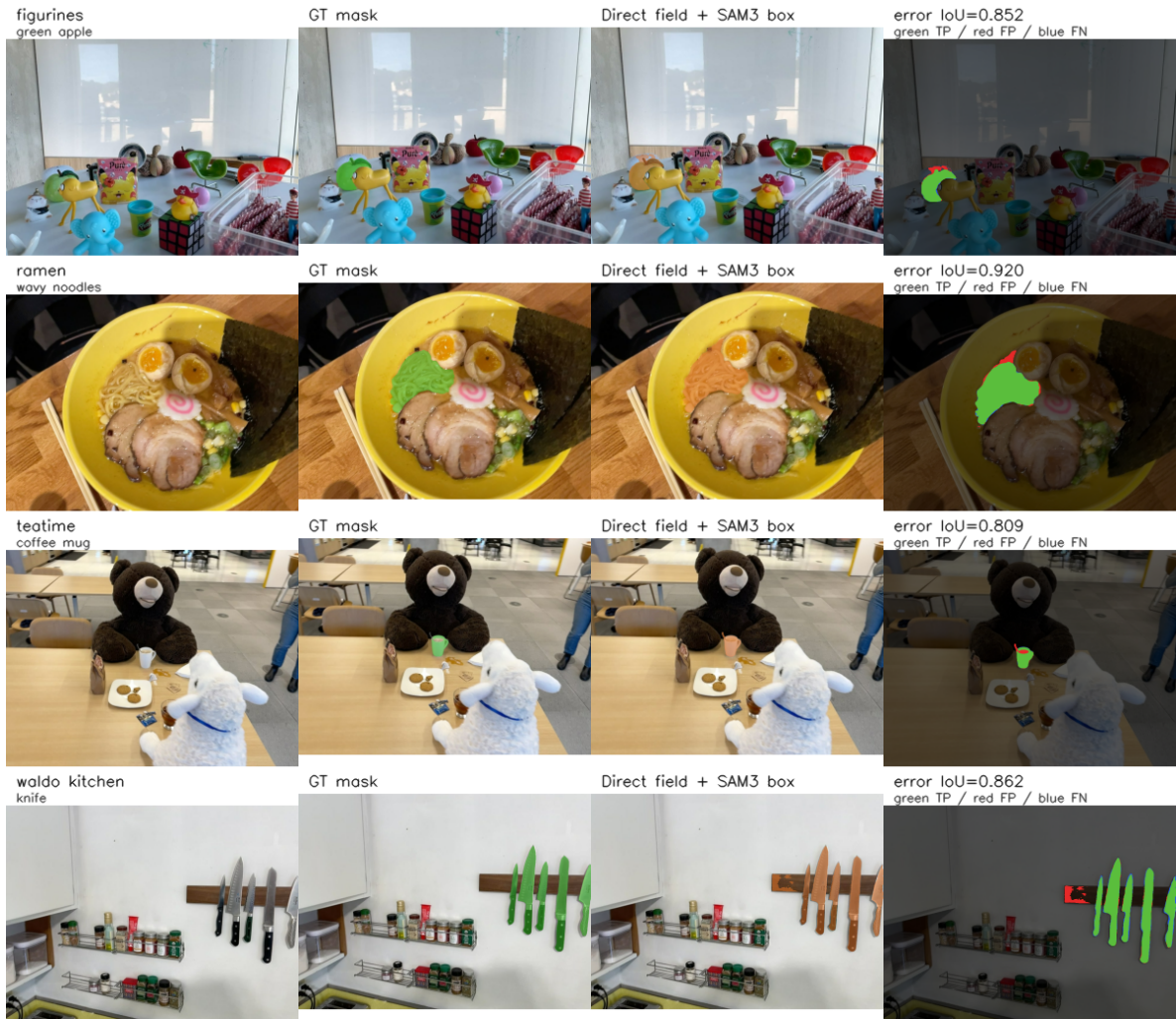
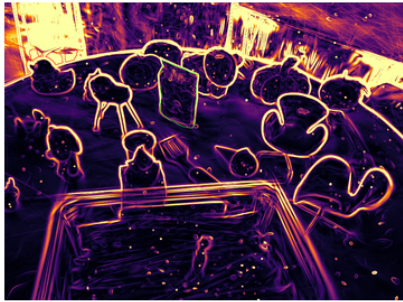


Fig. 3. Boundary-assisted direct-3D examples using the SAM3-box control. This variant illustrates a boundary diagnostic after compact primitive selection and is separate from the strict compact one-map ablation.



1. Figurines / pikachu | IoU 0.00, BE 1.00



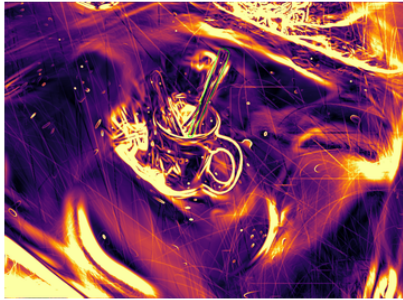
2. Ramen / napkin | IoU 0.00, BE 1.00



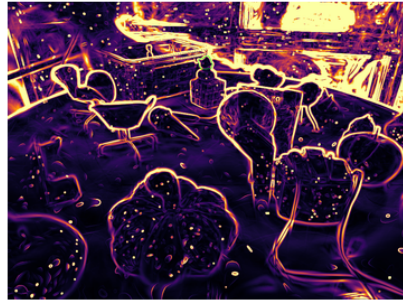
3. Teatime / yellow pouf | IoU 0.00, BE 1.00



4. Waldo Kitchen / knife | IoU 0.00, BE 1.00



5. Figurines / pirate hat | IoU 0.00, BE 1.00



6. Figurines / pirate hat | IoU 0.00, BE 1.00

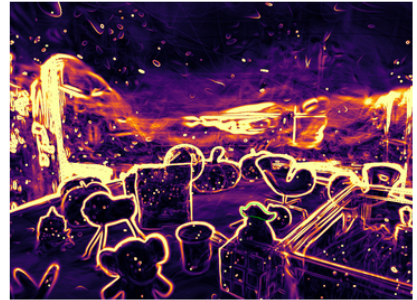


Fig. 4. Alpha/depth boundary cases used for geometric boundary analysis. These cases expose failure modes where low object support or depth discontinuities limit mask overlap.